GENERAL NOTES: UNLESS OTHERWISE SPECIFIED

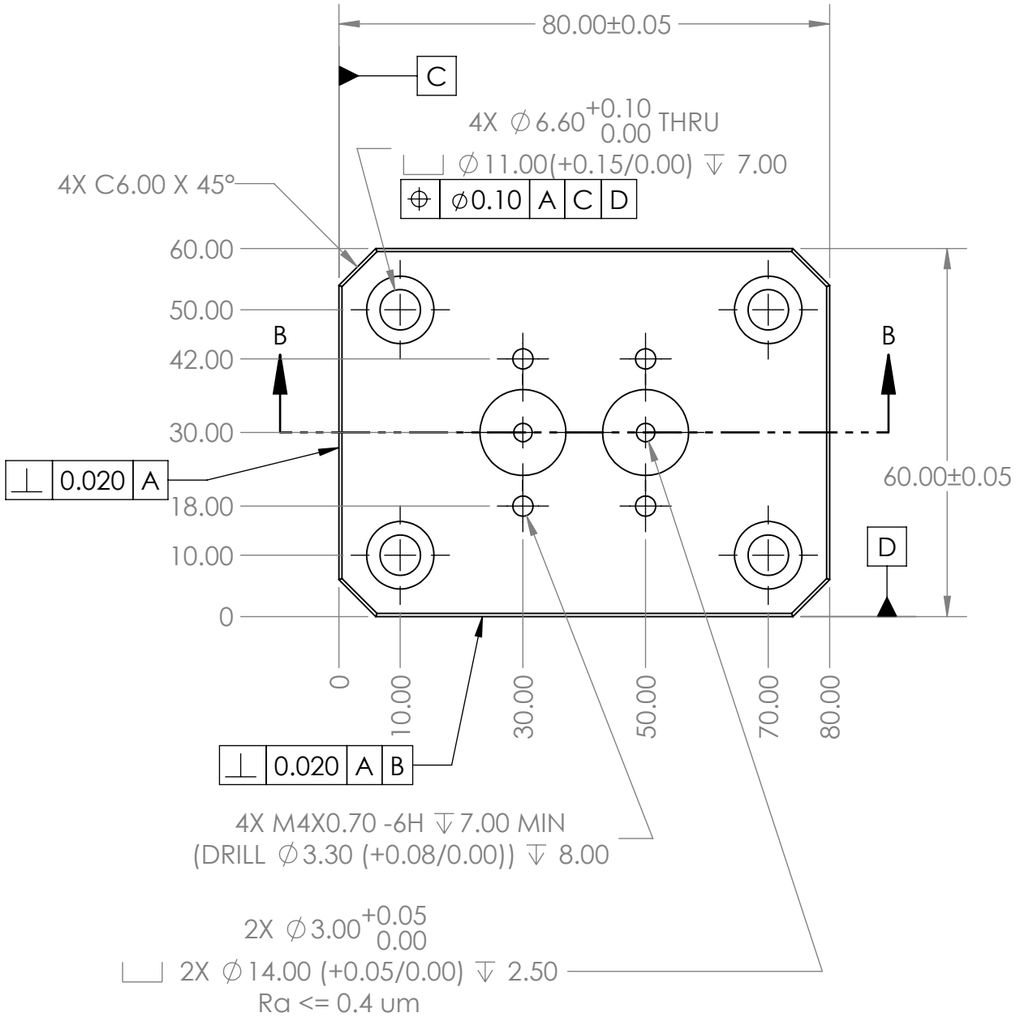
1. GENERAL & STANDARDS COMPLIANCE:
1.1 INTERPRET DRAWING IN ACCORDANCE WITH ASME Y14.5-2018 AND ISO 8015.
1.2 3D CAD DATASET (W3-AERO-VALVE-BLOCK-2.3.STEP) GOVERNS NOMINAL GEOMETRY FOR UNTOLERANCED SURFACES.
1.3 GENERAL TOLERANCES: LINEAR +/-0.05 mm, ANGULAR +/-0.2 DEG (PRECISION 5-AXIS CNC MACHINING).
1.4 THIRD ANGLE PROJECTION APPLIES THROUGHOUT. ALL UNITS IN MILLIMETERS.
1.5 COORDINATE REFERENCE FRAME: FIRST OCTANT CONVENTION (ALL POSITIVE X, Y, Z FROM DATUM ORIGIN C-D-A).

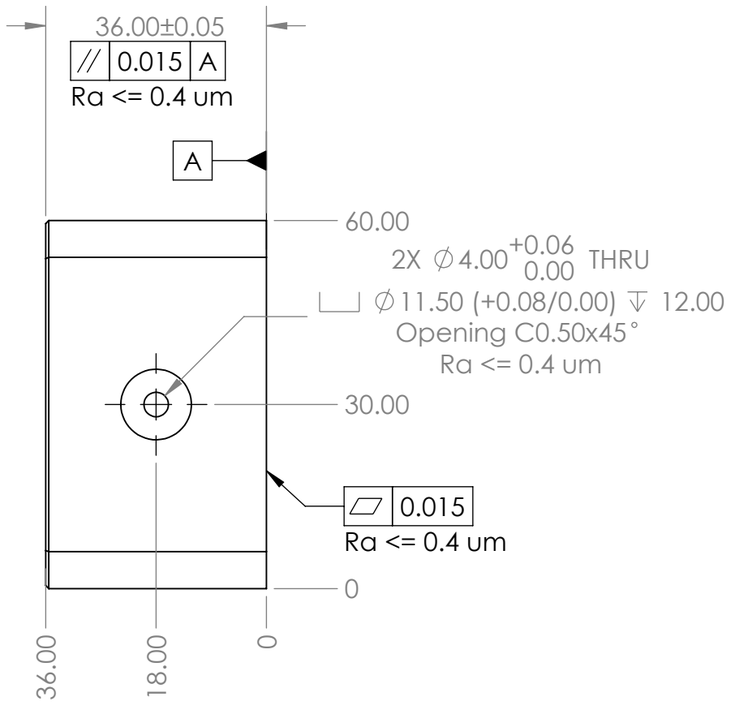
2. MATERIAL & AEROSPACE FLUID COMPATIBILITY:
2.1 MATERIAL OPTION A: FORGED AEROSPACE ALUMINUM ALLOY 7075-T7351 (AMS 4045 / ASTM B209 / EN AW-7075), OVERAGED FOR STRESS-CORROSION CRACKING RESISTANCE.
2.2 MATERIAL OPTION B: VACUUM INDUCTION MELTED NICKEL ALLOY INCONEL 718 (UNS N07718 / AMS 5662), HIGH TEMPERATURE AND CRYOGENIC COMPATIBLE.
2.3 100% SUBTRACTIVE 5-AXIS CNC MACHINING ONLY. NO ADDITIVE MANUFACTURING OR WELDED PLUGS ALLOWED TO ELIMINATE RISK OF PARTICULATE SHEDDING IN HYPERGOLIC / CRITICAL PROPULSION SYSTEMS.

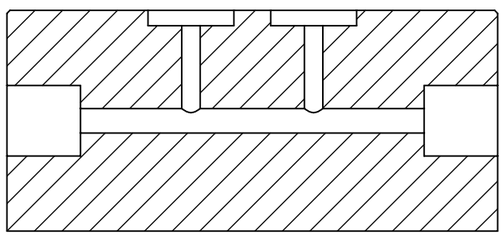
3. PRESSURE INTEGRITY & LEAKAGE TESTING:
3.1 OPERATING PRESSURE: 35.0 MPa (350 BAR).
3.2 PROOF PRESSURE TEST: 100% HYDROSTATIC PROOF TESTING AT 52.5 MPa (525 BAR, 1.5X PROOF) HELD FOR 10 MINUTES. ZERO PRESSURE DECAY OR PERMANENT PLASTIC DEFORMATION ALLOWED.
3.3 BURST PRESSURE SPECIFICATION: CALCULATED BURST PRESSURE >= 87.5 MPa (875 BAR, 2.5X SAFETY MARGIN).
3.4 HELIUM MASS SPECTROMETRY LEAK TEST AT 25.0 MPa: LEAK RATE < 1E-8 mbar·L/s (ZERO VISIBLE BUBBLES).

4. GD&T & SURFACE CONTROLS:
4.1 PRIMARY MOUNTING DATUM [A]: FLATNESS <= 0.015 mm, SURFACE FINISH Ra <= 0.8 MICRONS.
4.2 VALVE MATING INTERFACE [B]: FLATNESS <= 0.015 mm, PARALLEL TO [A] WITHIN 0.015 mm, Ra <= 0.4 MICRONS.
4.3 HIGH-PRESSURE INTERNAL PASSAGES (Ø4.00 mm AND Ø3.00 mm): INNER SURFACE FINISH Ra <= 0.4 MICRONS (EXTRUDE HONE / ABRASIVE FLOW POLISHED) TO MINIMIZE CAVITATION AND FLOW FRICTION.
4.4 METALLIC SEAL POCKETS (Ø14.00 mm x 2.50 mm DP): FLOOR FLATNESS <= 0.008 mm, Ra <= 0.4 MICRONS FOR METALLIC C-RING / CRUSH WASHER SEATING.

5. CLEANLINESS & PACKAGING:
5.1 COMPONENT MUST BE CLEANED PER IEST-STD-CC1246D LEVEL 100A (AEROSPACE PROPULSION GRADE CLEANLINESS).
5.2 100% FREE OF CUTTING FLUIDS, SWARF, AND PARTICULATE MATTER LARGER THAN 5 MICRONS.
5.3 VACUUM SEALED IN CLEANROOM PACKAGING WITH DESICCANT FOR FLIGHT SPARES STORAGE.







SECTION B-B

UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: Linear ±0.05 mm, Angular ±0.2° (5-Axis Precision CNC Milling)		FINISH: Type III Hard Anodize (MIL-A-8625, 25 µm thick for Al) / Native Passivated (ASTM A967 for Inconel)		DEBURR AND BREAK SHARP EDGES		DO NOT SCALE DRAWING	REVISION: A
	NAME	SIGNATURE	DATE			TITLE: AEROSPACE PROPULSION HIGH-PRESSURE DUAL-VALVE MANIFOLD BLOCK	
DRAWN							
CHK'D							
APPV'D							
MFG							
Q.A				MATERIAL: Option A: Aerospace Aluminum 7075-T7351 (AMS 4045 / ASTM B209) / Option B: Nickel Alloy Inconel 718 (AMS 5662)		DWG NO. W3-AERO-VALVE-BLOCK-2.3	A3
				WEIGHT: ~447.1 g (Al 7075) / ~1303.2 g (Inconel 718)		SCALE : NTS	SHEET : 1 OF 1